PantryParser: A Vegan Recipe Recommender System

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ABSTRACT

Finding recipes that match the ingredients available at home is a common challenge for home cooks, especially for those following vegan diets. Traditional recipe platforms often require users to search by dish name or cuisine, which can be inefficient and frustrating. PantryParser is a system designed to help users find vegan recipes based on the ingredients they have at home and answer questions about those recipes. Users can input a list of ingredients, and the system retrieves relevant recipes from a vegan dataset. The users can then ask questions about a choosen recipe and the system provides precise answers. By focusing on vegan recipes, the system also contributes to promoting healthier and more sustainable eating habits. PantryParser is build on a combination of ingredient preprocessing, a robust text-matching retrieval algorithm, and a fine-tuned extractive QA model. Evaluation on synthetic test cases showed that the system performs well: the correct recipe was ranked at the top in about 58% of cases and appeared in the top 10 almost 90% of the time. The QA component achieved an Exact Match of 83.4% and an F1 score of 92.46%, indicating high accuracy in extracting key information. Overall, the system shows that combining ingredient-based retrieval with question answering can provide an accessible and practical tool for vegan cooking.

CCS CONCEPTS

• **Computing methodologies → Information extraction**; **Natural language generation**; **Information extraction**; • **Information systems → Language models**; **Search interfaces**; **Question answering**; **Recommender systems**; **Information extraction**.

KEYWORDS

Natural Language Processing, Information Extraction, Question Answering, Recipe Retrieval, Ingredient Matching, Information Retrieval, Named Entity Recognition, Text Preprocessing, Vegan Recipes

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1 INTRODUCTION

Cooking is one of the most universal human activities and cooking at home has become more popular in recent years, yet finding the right recipe can still be surprisingly difficult [17, 22, 23]. Often, people end up browsing multiple websites, trying to figure out substitutions, or adjusting quantities manually. Most popular recipe platforms and websites expect users to start their search with the name of a dish or a broad cuisine category like "desserts" or "pasta." In practice, however, many home cooks begin with a very different question: "What can I make with the ingredients I already have at home?" [14]. This difference between how search interfaces are designed and how users actually think about cooking often leads to inefficiency, frustration and wasted ingredients. For example, a person with chickpeas, tomatoes, and spinach may not know what dishes they could make and a typical recipe site would not help unless they already had a dish in mind.

At the same time, there has been a significant increase in interest in plant-based and vegan diets [10, 11, 16]. Vegan cooking, which avoids all animal-based ingredients such as meat, dairy and eggs, can be challenging for beginners. It often requires substitutions, creative use of plant-based proteins, and familiarity with preparation methods that are less common in traditional cooking [1, 2]. A digital assistant that could help users find vegan recipes based on what they already have in their pantry could not only make cooking easier but also promote healthier eating habits and more environmentally sustainable food choices. This makes vegan recipe recommendation an interesting and socially meaningful point to work on [26].

From a technical perspective, this challenge falls within the scope of Natural Language Processing. Recipes are semi-structured texts that include ingredient lists, instructions and sometimes additional metadata like preparation time, cooking difficulty and user ratings [3]. Processing this type of data requires several NLP techniques. Simple tasks like normalization and tokenization are necessary for text comparison, while more advanced techniques such as fuzzy matching, named entity recognition, or question answering can allow a system to retrieve recipes that match user input and answer natural language questions about those recipes.

In this project, we introduce PantryParser, a prototype system that shows how NLP can be used for real-world recipe search. The system lets users enter ingredients through a simple console interface, then looks through a dataset of vegan recipes and ranks the most relevant dishes based on the ingredients provided. The system also has a question answering feature, so users can ask natural language questions about a recipe and get an answer automatically pulled from the recipe text.

To build PantryParser, we used a publicly available vegan recipe dataset that includes both ingredients and instructions. We implemented a preprocessing pipeline to clean and standardize the ingredients, and a scoring system that ranks recipes based on how well they match the user's input. For answering questions about the recipes, we used a fine-tuned extractive question answering model. Evaluations on synthetic test cases showed that the system can reliably find relevant recipes and provide accurate answers, even when users only enter partial ingredient lists. This paper describes the methods used to build PantryParser, presents the results of the evaluation, and discusses the system's strengths, limitations, and potential future improvements. By combining recipe retrieval and question answering into a single system, this system demonstrates how NLP can make vegan cooking easier and more convenient for everyone.

Overall, this project adds to the development of smart cooking assistants and digital tools for meal planning. It shows how NLP techniques can be used outside of typical text analysis to tackle a real, everyday problem. By focusing on vegan recipes, it also looks at a growing interest in plant-based cooking, helping users make meals efficiently while keeping environmental impact in mind. Through this work, we hope to highlight both the technical and user-focused challenges of building recipe recommendation systems and suggest ideas for future improvements and research.

2 RELATED WORK

Food recommendation systems have grown largely over the last decade and are becoming increasingly important as more people turn to digital tools for cooking inspiration and meal planning. These systems don't just suggest meals, they can help users make healthier choices and explore new types of cuisine. Research shows that while predicting user preferences for food is possible, it is still more challenging than in other domains like movies or music, with standard methods often producing only modest results. Context, such as social situations or dietary needs, plays a significant role in food choices, and incorporating this information can improve recommendations. Nutritional health is also a growing focus, with researchers experimenting with ways to guide users toward healthier meals, though no single approach has emerged as the best yet. Evaluation remains another challenge, as most studies rely on offline or proprietary datasets rather than testing systems in real-world scenarios [24, 26].

Food recommendation systems can play a key role in helping people maintain a healthy diet by not just suggesting individual recipes, but by creating balanced meal plans. Research has shown that it is possible to algorithmically combine recipes into daily plans that meet nutritional guidelines, while taking into account users' food preferences. These studies also highlight challenges, such as determining how many recommendations are needed and how to ensure that meal plans are genuinely balanced from a health perspective [6, 20, 27].

Research also shows that people often struggle to judge the nutritional content of recipes, with choices frequently influenced by visual cues and other misleading signals rather than actual healthiness. Studies have shown that by using machine learning and recipe data, like images and metadata, recommendation systems can predict user preferences and even nudge them towards healthier options without reducing enjoyment [8].

Lifestyle-related illnesses like diabetes and obesity are a growing concern, but good nutrition can help prevent or even reverse them. Incorporating healthiness into food recommendations has the potential to help people make better eating choices, especially as more users turn to online platforms for cooking inspiration. Food recommender systems have the potential to support healthier eating habits, but traditional approaches often focus only on predicting what users like, which can unintentionally reinforce preferences for high-fat or high-calorie meals. Research suggests that by reformulating the recommendation process to include nutritional considerations alongside user preferences, these systems can guide people toward more balanced diets [7, 13].

Building on this idea, AI can also support more specific dietary goals, such as adopting plant-based diets, which are becoming increasingly important not only for health but also for sustainability. Many people, find it challenging to adapt traditional recipes to be vegan. Research on systems like Veganaizer demonstrates that machine learning models trained on large collections of vegan and omnivorous recipes can suggest suitable plant-based substitutes for non-vegan ingredients. By analyzing ingredient embeddings, such systems can recommend replacements that maintain the original flavor and texture, although challenges remain, such as handling ingredients that serve multiple functions in different recipes or considering cooking techniques that affect usability [15].

On the same note reducing food waste is a major global concern, and one practical way to address it is by helping people make meals with the ingredients they already have at home. Systems like RecipeIS show that using AI to recognize ingredients, through methods like image recognition with convolutional neural networks, can guide users in discovering recipes they might not have thought of. The model can identify ingredients accurately and then fetch recipes from an external database or API, making recommendations more personalized and convenient [21].

Freyne demonstrated that recipe recommendation systems can be effective if they balance accuracy, coverage, and the effort required from users to provide input. Even simple content-based approaches that break down recipes into individual ingredients can outperform more complex collaborative filtering when data is sparse [9]. Morol et al. extended this idea by showing that deep learning can be used to recognize food ingredients from images, which can then drive recipe recommendations. They trained a convolutional neural network on a custom dataset of over 9,800 images across 32 ingredient classes, achieving an impressive 94% accuracy in ingredient detection. By combining this ingredient recognition with a recommendation algorithm, their system could suggest recipes based on the identified ingredients, making it easier for both beginners and experienced cooks to decide what to prepare [18].

Research shows that these systems have evolved from simple content-based and collaborative filtering methods to approaches that use graph models and deep learning, which improve both accuracy and reliability. Nutrition- and health-focused methods, in particular, take into account the connection between users' health and the nutritional content of recipes, offering more satisfying and beneficial recommendations. However, challenges remain, including data collection, accommodating diverse user needs, and integrating nutritional knowledge effectively [9, 29]. Similar research shows that recommendation systems can help by suggesting recipes based on the ingredients available, using techniques like content-based filtering and ingredient co-occurrence analysis. These systems can save time, reduce the effort of searching online, and even allow users to save or create their own recipes for later use [5, 28].

More recent research has shown that recipe recommendation can be improved by looking beyond simple content-based or collaborative filtering approaches and considering the relationships between users, recipes, and ingredients. For example, RecipeRec models these connections using a large-scale user-recipe-ingredient graph and a heterogeneous graph neural network, capturing both recipe content and relational structure to make more accurate recommendations. The system even uses contrastive learning to enhance the information extracted from the graph, outperforming traditional methods [19, 25].

PantryParser applies a similar principle in a simpler, user-friendly way and builds on all these insights by allowing users to input ingredients and receive vegan recipe suggestions while also supporting interactive questions about the recipes. By combining ingredient-based recommendations with user-friendly interaction, the system aims to make cooking more accessible, personalized, and engaging, while addressing some of the key challenges highlighted in current research on food recommender systems.

3 METHOD

3.1 Dataset

For this project, we decided to use a publicly available dataset instead of scraping recipes from websites. Scraping can cause legal issues and also makes it hard for others to reproduce results, so we decided to use the **Kasaf/version2-vegan-recipes dataset** [12] hosted on HuggingFace. This dataset contains vegan recipes in two main columns: "title" and "combined_plain". The combined_plain field merges all the important information Recipe Title, Ingredient, Method, Environmental Impact and Nutritional Information into plain text.

Using this pre-existing dataset had several advantages. First, it allowed us to start building and testing the system quickly without spending time collecting and cleaning web data. Second, it ensures that our work can be replicated by others, which is important for reproducibility in research projects. Third, because it is already curated for vegan recipes, it aligns well with the focus of PantryParser, helping us avoid the need to filter out non-vegan recipes manually. However, using this dataset also comes with limitations. Since the ingredient and instruction fields are combined, we had to rely on heuristic methods to separate them and some metadata like prep time, difficulty, and ratings are not available.

3.2 Preprocessing

The recipes in the dataset are provided in unstructured text, so preprocessing was an important first step. Without preprocessing, it would have been difficult for the system to understand and match ingredients or answer questions accurately. The preprocessing pipeline consisted of several steps:

3.2.1 Normalization and Extraction. Instead of basic normalization, we used regular expressions (Regex) within a custom function (`extract_ingredients_and_title`) to achieve surgical cleaning:

(1) We first pulled out the Recipe Title.
(2) For ingredients, we applied a Regex pattern to strip away all quantities, units and descriptive adjectives from each line.
(3) The remaining, core ingredient name was then simplified to only the first few important words (max 3 words, each longer than 2 characters).
(4) These clean, simplified ingredients were stored in a set (`ingredients_set`) to ensure only unique items were available for searching.

Any recipes that resulted in an empty ingredient set were filtered out. We also saved the original index as the **`recipe_id`** for evaluation tracking. Preprocessing was essential not only for functionality but also for performance. By reducing noise and structuring the text, the system could provide more accurate matches and answers and it made string comparisons more consistent.

3.3 Recipe Retrieval

The core of PantryParser is the ingredient-based recipe retrieval in the `search_recipes` function, which matches the user's input to the recipe's `ingredients_set`. Users enter ingredients (e.g., "tomato, onion, basil") and the more ingredients are in a recipe, the higher the recipe gets shown in the list. Therefore we implemented a robust scoring function (`score_recipe_robust`) that goes beyond simple string containment by using a two-pronged matching strategy:

- **Full Token Match (Option A)**: Checks if an entire user ingredient (e.g., "sweet potato") exactly matches an ingredient in the recipe's set.
- **Word-Level Match (Option B)**: If Option A fails, it checks if any individual word from the user's input matches any word in the recipe's ingredient list. This catches partial matches (e.g., input "black beans" matching a recipe ingredient of "black bean").

Recipes are scored based on two factors from this matching:

- **Matching Count**: The total number of unique user ingredients matched.
- **Coverage Score**: $\frac{\text{Matching Count}}{\text{Total Recipe Ingredients}}$. This score prioritizes recipes that use a high proportion of the ingredients provided.

The recipes are ranked by the highest matching count first, and then by the highest coverage score as a tie-breaker. The top K=10 results are returned.This approach is simple but effective. It captures exact matches and minor variations (like "chickpeas" vs. "chickpea") while keeping computation lightweight. Future improvements could include semantic embeddings to recognize synonyms and ingredient substitutions more robustly.

3.4 Question Answering

For the QA model, we chose the `deepset/roberta-base-squad2` [4] model because it offers an ideal balance between high accuracy and moderate computational efficiency for our specific task of extractive question answering. The model is based on the RoBERTa architecture, which is an improved variant of the original BERT model and achieves superior language comprehension performance through longer training on a larger amount of data. In particular, this version has been fine-tuned on the SQuAD 2.0 dataset, the gold standard for filtering answers from texts. This specialization ensures that the model not only knows where the information is located, but can also accurately extract the exact phrase to provide reliable quantity information to the user. This was instrumental in achieving the high EM and F1 scores shown in the results.

Users can ask questions like "How long does this take?" or "How many servings?". Instead of passing the entire recipe text, we implemented a context-routing strategy to improve accuracy and reduce inference time:

- If the question contained keywords like "how much" or "quantity", only the **Ingredients** section was provided as context.
- If the question contained keywords like "how long" or "cook", only the **Method** section was provided as context.
- For all other questions, the entire recipe text was used.

3.5 Evaluation

We evaluated PantryParser using two separate test situations, which are automatically generated from the dataset itself.

3.5.1 Recipe Retrieval Evaluation. We assessed the retrieval system's quality by running the `evaluate_retrieval_synthetic` function. This is a common evaluation where the system must find the correct recipe when queried with a subset of its own ingredients.

- **Test Setup**: We ran 500 trials. In each trial, we selected a random recipe and queried the system with a subset of 4 of its ingredients.
- **Metrics**: We calculated standard ranking metrics:
  - **Mean Reciprocal Rank**: This score heavily penalizes results that appear lower in the ranked list.
  - **Recall@k**: This measures the percentage of trials where the correct recipe appeared within the top k results. We reported scores for R@1, R@5 and R@10.

3.5.2 Question Answering Evaluation. We tested the QA model's ability to extract quantities, as this is a common and verifiable task.

- **Test Setup**: We ran the `evaluate_qa_synthetic` function to generate **200 test cases**. This involved using Regex to find pairs of quantities and ingredients (e.g., "300g flour"), formulate a question (e.g., "How much flour is needed?"), and set the quantity as the `ground_truth_answer`.
- **Metrics**: We used the standard **SQuAD** evaluation metrics:
  - **Exact Match (EM)**: The percentage of predictions that perfectly match the ground truth answer.
  - **F1 Score**: A more forgiving metric based on the overlap of tokens between the predicted and actual answers.

4 RESULTS

4.1 Retrieval Performance

The ingredient matching part of PantryParser performed reasonably well, confirming that even simple text-based approaches can bring back useful and relevant recipe results. The robust performance of the `score_recipe_robust` was quantitatively assessed using a custom, synthetic test set. The idea was to see if the system could still find the original recipe when the search only included three randomly chosen ingredients from that recipe. The results, summarized in Table 1, suggest that the system is both accurate and effective at recipe retrieval.

The **Mean Reciprocal Rank** was 0.6842, which indicates solid performance. In practice, this means the system usually places the correct recipe near the top of the results list, so users don't need to look very far to find it.

The **Recall@k** scores are also quite strong:

- A **Recall@1** of 0.5780 means that in about 57.8% of the queries, the correct recipe appeared at the very top of the results. This highlights the effectiveness of the initial ranking.
- Looking further, the top 5 results (**Recall@5** = 0.8180) and top 10 results (**Recall@10** = 0.8920) almost always included a relevant recipe. This demonstrates that the embedding and indexing strategy is highly reliable for matching user ingredients to recipes.

4.2 Question Answering (QA) Performance

Next, we evaluated the QA component, which uses the retrieved recipes to answer specific questions about them. A total of 500 test cases were run, and the results, measured with standard evaluation metrics, are summarized in Table 2.

The QA component achieved strong results. The Exact Match (EM) score was 83.4%, meaning that in more than four-fifths of the cases, the system's answers were exact, word-for-word matches with the ground truth. This is a very precise result and suggests the model is highly accurate at extracting facts.

The F1 score was even higher at 92.46%. Unlike EM, F1 allows for minor wording differences as long as the key information is captured. The fact that the F1 score is close to 100% shows that even when the system's answers were not perfect matches, they still included all of the essential information.

4.3 Discussion of Findings

The results show that the system works quite well overall. For example, the Mean Reciprocal Rank of 0.6842 and Recall@1 of 0.5780 suggest that the correct recipes usually show up near the top of the list, which means the scoring method and simplification step are doing their job. The higher Recall@5 (0.8180) and Recall@10 (0.8920) scores make this even clearer, since they show that a relevant recipe almost always appears within the first few results. That said, it's worth noting that these numbers come from a synthetic test set designed for evaluation. Real user queries are often less clean and more unpredictable, so the real-world performance might be a bit lower. Still, these strong baseline results highlight that the `score_recipe_robust` algorithm is a solid starting point for building a reliable recipe retrieval system.

4.4 Discussion of Findings

The results indicate that the system performs effectively across the evaluated metrics. The Mean Reciprocal Rank of 0.6842 and Recall@1 of 0.5780 show that the correct recipes are frequently

Table 1: Retrieval Performance Metrics (N=500 Trials)

Metric	Value	Interpretation
Mean Reciprocal Rank	0.6842	How high up the first correct result appears
Recall@1	0.5780	Correct recipe is the #1 suggestion
Recall@5	0.8180	Correct recipe is in the top 5
Recall@10	0.8920	Correct recipe is in the top 10

Table 2: Question Answering Performance Metrics (N=500 Trials)

Metric	Value (%)	Interpretation
Exact Match	83.4000	Answer is a perfect match to the reference
F1 Score	92.4570	Measures token overlap between prediction and reference

ranked near the top, demonstrating the usefulness of the scoring method and simplification step. The strong Recall@5 (0.8180) and Recall@10 (0.8920) values further confirm the system's ability to consistently identify relevant recipes, even when queries are partial or noisy. It should be noted, however, that these results are based on a controlled synthetic test set, meaning that real-world queries, which often are more varied and less precise, may lead to somewhat lower performance. Nonetheless, the strong baseline established here highlights the `score_recipe_robust` algorithm as a reliable foundation for a recipe retrieval system.

4.5 Conclusion on Performance

Overall, the system's performance is highly positive, with strong results across multiple areas. The high recall scores show that PantryParser can reliably find the right recipes, while the high EM and F1 scores demonstrate that it can accurately pull the correct information from them. In practice, we also saw that it performs especially well for straightforward questions with clear numerical answers, like "How many servings?", and its ingredient matching is very precise, ensuring that users typically receive relevant vegan recipes. At the same time, some answers were vague or cut off—for instance, when asked about cooking time, the model might reply with "about half" instead of "about half an hour." These issues highlight the limits of purely extractive QA models and the importance of hybrid approaches that combine statistical methods with rule-based constraints. Taken together, the results show that this architecture is clearly effective for recipe retrieval and question answering, but also that improvements are needed before the system can be considered fully reliable for real-world use.

5 DISCUSSION

The results of this project confirm that even relatively simple NLP methods can be used to build a functional and useful recipe retrieval system. PantryParser works quite well overall, but there are also some clear areas where it can be improved. For the recipe retrieval, the scoring system based on ingredient overlap did a solid job and. The correct recipe was often ranked very high, with Recall@10 close to 90%. This means that in most cases, the recipe a user is looking for would show up within the first few results and the high Mean Reciprocal Rank showed that the system efficiently pushes the most relevant recipes to the top, which is practical and user-friendly. These results show that focusing on matching recipes by ingredient count and how well they cover the recipe helps make it easier for users to find what they're looking for. The fact that even a relatively simple text-matching approach achieved this shows that sometimes lightweight methods can go a long way.

At the same time, the system is still limited by the way it matches text. Ingredients can be expressed in multiple ways, for example, "chickpea" vs. "garbanzo bean" or "plant-based milk" vs. "soy milk" wouldn't count as the same ingredient, even though they are. The system often fails to capture these semantic connections and as a result, relevant recipes may be overlooked. This explains why Recall@1 was not higher. Only about 58% of the time was the correct recipe ranked at the very top. In a real-world setting, this could mean that users sometimes need to scroll or adjust their input if their wording doesn't exactly match what's in the dataset. One good improvement for the future could be using embedding-based methods to better understand the actual meaning of ingredients. This would probably help catch more matches, even when synonyms are used, and make the recommendations work better overall.

The question answering part performed quite good too, with high Exact Match and F1 scores. This demonstrates that PantryParser is reliably pulling out the necessary and specific details like serving sizes or ingredient amounts, even if formatting occasionally differs. We think this worked well partly because we split up the recipe text so the model only looked at the "Ingredients" section for quantity questions. That way there was less noise and the answers came out more accurate. However, there were some cases where the answers were incomplete or not very clear, such as giving "about half" instead of "about half an hour." Instructions often mention times, amounts, or temperatures in less exact ways, like "bake for about half an hour" or "add a pinch of salt." These cases require some reasoning about time and quantities, which standard models still tend to struggle with. So these small mistakes make sense because the model is only designed to extract text spans, not rewrite or add context. In real use, this could confuse some users if they are expecting full, natural answers.

It's also worth pointing out that the tests we ran were based on synthetic cases created from the dataset. While this is useful for measuring performance in a controlled way, it doesn't fully reflect how real users would interact with the system. Actual user questions are often less direct, more varied or noisy. So, the results we report here are more of a "best case scenario," and the actual performance in the real world might be a bit lower. Nevertheless, overall the experiments suggest that PantryParser has a strong foundation. It's reliable enough to be useful, especially for straightforward queries, but it also has some gaps that would need to be addressed before it could be considered a robust everyday tool.

Another limitation of the current system is its reliance on a single, pre-existing dataset where fields are aggregated. While this made implementation faster and avoided potential legal or reproducibility issues with web scraping, it also limits the variety and detail of information available for ranking and recommendations. Adding more metadata, like ratings, prep time, difficulty, or nutritional info, could make the system more precise and personalized (e.g. suggesting quick meals, high-protein recipes, or low-calorie options).

Finally, the system also hasn't been tested with real users yet, so it's unclear how well it works in a real kitchen. Real-world testing with actual users is an important direction for future exploration. User studies could reveal usability issues, gaps in ingredient recognition, and unexpected preferences, and they could help guide features like substitutions, dietary flexibility, or recipe diversity. Right now, every user gets the same results for the same input, which limits engagement, but future improvements could track preferences, use feedback loops, or apply collaborative filtering to make recommendations smarter and more tailored. Combining richer data, real-world testing, and personalization would make the system more practical, responsive, and user-friendly over time.

6 CONCLUSION & FUTURE WORK

In this project, we built PantryParser, a system that helps users find vegan recipes based on the ingredients they have and then ask questions about those recipes. This project shows that it is possible to use NLP techniques in a practical, user-focused way for recipe retrieval. The system combines two main parts: an ingredient-based retrieval system and a question answering model, making it more interactive than a simple search tool. Our evaluation showed that both parts performed reasonably well. The retrieval system was usually able to rank the correct recipe near the top of the results, and the QA model was able to return precise answers with high accuracy.

Users can find recipes that include most of their ingredients, and the system avoids completely irrelevant suggestions. However there are still some challenges especially for recipes that use synonyms or less common ingredient names. The retrieval system depends on exact text matches, which makes it less effective when the user and the dataset use different words for the same thing. The QA system, while accurate, can sometimes give answers that are technically correct but incomplete or awkward. Also, since we tested everything using automatically generated queries, there's still some uncertainty about how the system would hold up with real users. Even with these limitations, PantryParser shows that simple NLP tools can still be useful for everyday tasks like cooking and planning meals.

Focusing on vegan recipes shows another interesting aspect of the system: NLP can help encourage healthier and more sustainable eating. By making it easier to find creative plant-based meals, the system supports these broader goals and tackles the technical challenge of making sure recommendations stick to specific diet rules. This mix of practical use and social impact makes the system an interesting and useful experiment.

For future work, there are several areas for improvement and further research. One would be to use semantic embeddings so that the system can understand synonyms and substitutions. Another would be to move beyond purely extractive question answering and try a generative approach, which could give smoother, more natural answers. Finally, testing the system with actual users would give better feedback on what works well and what needs fixing. There is also potential for personalization: tracking users' favorite ingredients, cuisines, or cooking styles could make the system smarter over time.

In summary, this project demonstrates that combining relatively simple NLP techniques with thoughtful design can produce a system that is both functional and socially meaningful. While there are limitations to address, this project provides a strong foundation. With continued development, it could become a genuinely helpful tool for anyone looking to cook creatively and sustainably with the ingredients they already have.

REFERENCES

[1] Alexandra Alcorta, Adrià Porta, Amparo Tárrega, María Dolores Alvarez, and M. Pilar Vaquero. 2021. Foods for Plant-Based Diets: Challenges and Innovations. *Foods* 10, 2 (2021). https://doi.org/10.3390/foods10020293

[2] Mathilde Beck, John Harvey, and Christina Trauth. 2017. Veganism: Motivations and Difficulties. *Methodology* 1, 5 (2017).

[3] Michał Bień, Michał Gilski, Martyna Maciejewska, Wojciech Taisner, Dawid Wisniewski, and Agnieszka Lawrynowicz. 2020. RecipeNLG: A cooking recipes dataset for semi-structured text generation. In *Proceedings of the 13th international conference on natural language generation.* 22–28.

[4] Branden Chan, Timo Möller, Malte Pietsch, and Tanay Soni. 2024. deepset/roberta-base-squad2. https://huggingface.co/deepset/roberta-base-squad2 Accessed: 2025-09-30.

[5] Soundarya Desai, Ms Pooja Patil, Mr Pratik Shinde, Mr Azhar Sayyed, and Rohini Bhosale. 2019. Recipe recommendation based on ingredients using machine learning. *International Journal of Advanced Research in Computer and Communication Engineering* 8, 3 (2019).

[6] David Elsweiler and Morgan Harvey. 2015. Towards Automatic Meal Plan Recommendations for Balanced Nutrition. In *Proceedings of the 9th ACM Conference on Recommender Systems* (Vienna, Austria) *(RecSys '15)*. Association for Computing Machinery, New York, NY, USA, 313–316. https://doi.org/10.1145/2792838.2799665

[7] David Elsweiler, Morgan Harvey, Bernd Ludwig, and Alan Said. 2015. Bringing the" healthy" into Food Recommenders. *Dmrs* 1533 (2015), 33–36.

[8] David Elsweiler, Christoph Trattner, and Morgan Harvey. 2017. Exploiting Food Choice Biases for Healthier Recipe Recommendation *(SIGIR '17)*. Association for Computing Machinery, New York, NY, USA, 575–584. https://doi.org/10.1145/3077136.3080826

[9] Jill Freyne and Shlomo Berkovsky. 2010. Intelligent food planning: personalized recipe recommendation. In *Proceedings of the 15th International Conference on Intelligent User Interfaces* (Hong Kong, China) *(IUI '10)*. Association for Computing Machinery, New York, NY, USA, 321–324. https://doi.org/10.1145/1719970.1720021

[10] Nina Gheihman. 2021. Veganism as a lifestyle movement. *Sociology compass* 15, 5 (2021), e12877.

[11] Mikołaj Kamiński, Karolina Skonieczna-Żydecka, Jan Krzysztof Nowak, and Ewa Stachowska. 2020. Global and local diet popularity rankings, their secular trends, and seasonal variation in Google Trends data. *Nutrition* 79-80 (2020), 110759. https://doi.org/10.1016/j.nut.2020.110759

[12] Kasaf. 2023. Version2 Vegan Recipes. https://huggingface.co/datasets/Kasaf/version2-vegan-recipes Accessed: 2025-09-30.

[13] Mansura Khan, Ellen Rushe, Barry Smyth, and David Coyle. 2019. Personalized, Health-Aware Recipe Recommendation: An Ensemble Topic Modeling Based Approach. *ACM SIGIR Forum* (09 2019).

[14] Marian Knopp. 2011. Information needs, preferences, and behaviors of home cooks. *Library and Information Research* 35, 109 (2011), 40–54.

[15] Dennis Lawo, Lukas Böhm, and Gunnar Stevens. 2020. Veganaizer: AI-assisted ingredient substitution. *University of Siegen: Siegen, Germany* (2020).

[16] Dennis Lawo, Katharina Litz, Christina Gromov, Hannah Schwärzer, and Gunnar Stevens. 2019. Going Vegan: The Use of digital Media in vegan Diet Transition. In *Proceedings of Mensch Und Computer 2019* (Hamburg, Germany) *(MuC '19)*. Association for Computing Machinery, New York, NY, USA, 503–507. https://doi.org/10.1145/3340764.3344447

[17] Harold McGee. 2007. *On food and cooking: the science and lore of the kitchen.* Simon and Schuster.

[18] Md. Kishor Morol, Md. Shafaat Jamil Rokon, Ishra Binte Hasan, A. M. Saif, Rafid Hussain Khan, and Shuvra Smaran Das. 2022. Food Recipe Recommendation Based on Ingredients Detection Using Deep Learning. In *Proceedings of the 2nd International Conference on Computing Advancements* (Dhaka, Bangladesh) *(ICCA '22)*. Association for Computing Machinery, New York, NY, USA, 191–198. https://doi.org/10.1145/3542954.3542983

[19] Ruiqi Ouyang, Haodong Huang, Weihua Ou, and Qilong Liu. 2024. Multimodal Recipe Recommendation with Heterogeneous Graph Neural Networks. *Electronics* 13, 16 (2024). https://doi.org/10.3390/electronics13163283

[20] Sara Ozeki, Masaaki Kotera, Katushiko Ishiguro, Taichi Nishimura, and Keita Higuchi. 2022. Recipe Recommendation for Balancing Ingredient Preference and Daily Nutrients *(CEA++ '22)*. Association for Computing Machinery, New York, NY, USA, 11–19. https://doi.org/10.1145/3552485.3554941

[21] Miguel Simões Rodrigues, Filipe Fidalgo, and Ângela Oliveira. 2023. RecipeIS—Recipe Recommendation System Based on Recognition of Food Ingredients. *Applied Sciences* 13, 13 (2023). https://doi.org/10.3390/app13137880

[22] Frances Short. 2006. *Kitchen secrets: The meaning of cooking in everyday life.* Berg.

[23] Dean Simmons and Gwen E Chapman. 2012. The significance of home cooking within families. *British Food Journal* 114, 8 (2012), 1184–1195.

[24] Pradeep Kumar Singh, Pijush Kanti Dutta Pramanik, Avick Kumar Dey, and Prasenjit Choudhury. 2021. Recommender systems: an overview, research trends, and future directions. *International Journal of Business and Systems Research* 15, 1 (2021), 14–52.

[25] Yijun Tian, Chuxu Zhang, Zhichun Guo, Chao Huang, Ronald Metoyer, and Nitesh V. Chawla. 2022. RecipeRec: A Heterogeneous Graph Learning Model for Recipe Recommendation. arXiv:2205.14005 [cs.IR] https://arxiv.org/abs/2205.14005

[26] Christoph Trattner and David Elsweiler. 2017. Food Recommender Systems: Important Contributions, Challenges and Future Research Directions. arXiv:1711.02760 [cs.IR] https://arxiv.org/abs/1711.02760

[27] Christoph Trattner and David Elsweiler. 2017. Investigating the Healthiness of Internet-Sourced Recipes: Implications for Meal Planning and Recommender Systems. In *Proceedings of the 26th International Conference on World Wide Web* (Perth, Australia) *(WWW '17)*. International World Wide Web Conferences Steering Committee, Republic and Canton of Geneva, CHE, 489–498. https://doi.org/10.1145/3038912.3052573

[28] MB Vivek, N Manju, and MB Vijay. 2017. Machine learning based food recipe recommendation system. In *Proceedings of International Conference on Cognition and Recognition: ICCR 2016*. Springer, 11–19.

[29] Yao Wu, Ling Kang, Quan Guo, and Lei Zhao. 2025. A Review of Recipe Recommendation Methods. *IEEE Access* (2025).